

K210-Machine code following

K210-Machine code following

1. Opening Notes
2. Experimental Preparation
 - The relationship between the 4 motor interfaces and the car is as follows:
 - Hardware wiring
 - Overall wiring
 - Wiring pins
3. Key code analysis
4. Experimental operation
5. Experimental phenomenon

1. Opening Notes

Please read the "Introduction to Motors and Usage" in the information of the four-way motor driver board first to understand the motor parameters, wiring methods, and power supply voltage you are currently using. To avoid burning the motherboard or motor.

Motor: The case and code take the 310 motor of our store as an example.

2. Experimental Preparation

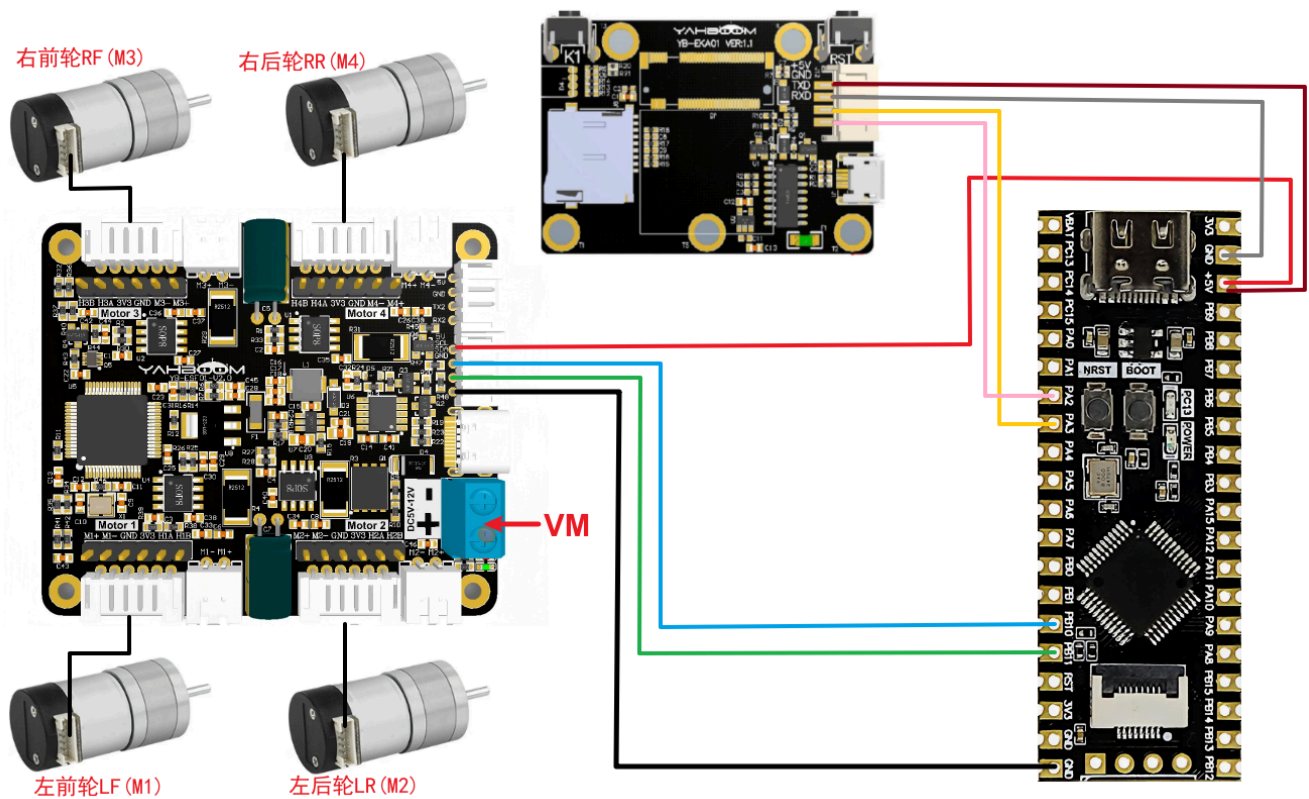
National Race Chassis V2 Four-wheel Drive Version, 4*310 Motors, 7.4V Lithium Battery, K210 Vision Module, STM32F103C8T6 Core Board.

The relationship between the 4 motor interfaces and the car is as follows:

- M1 -> upper left motor (left front wheel of the car)
- M2 -> lower left motor (left rear wheel of the car)
- M3 -> upper right motor (right front wheel of the car)
- M4 -> lower right motor (right rear wheel of the car)

Hardware wiring

Overall wiring



Wiring pins

Four-way motor driver board	STM32C8T6
5V	5V
GND	GND
SCL	PB10
SDA	PB11

Take M1 motor as an example below, and other motors are similar.

Motor	Four-way motor driver board (Motor)
M2	M1-
VCC	3V3
A	H1A
B	H1B
GND	GND
M1	M1+

K210	STM32C8T6
5V	5V

K210	STM32C8T6
GND	GND
TXD	PA3
RXD	PA2

3. Key code analysis

- bsp.c

```
#include "bsp.h"
void BSP_init(void)
{
    SystemInit();
    NVIC_PriorityGroupConfig(NVIC_PriorityGroup_2);

    delay_init();
    // delay_ms(1000); //等待红外稳定    wait for infrared to stabilize

    USART1_init(115200);
    USART2_init(115200); //使用串口2 接收红外    Use serial port 2 to receive infrared
    RGB_Init();
    IIC_Motor_Init(); //四路电机通信初始化    Four-way motor communication initialization

    //放到最后才生效，不然还是无法正常使用    It will take effect at the end, otherwise it will
    not work properly.
    RCC_APB2PeriphClockCmd(RCC_APB2Periph_AFIO, ENABLE);
    GPIO_PinRemapConfig(GPIO_Remap_SWJ_JTAGDisable, ENABLE); //禁用jlink 只用SWD调试口, PA15、
    PB3、4做普通IO    Disable jlink and use only SWD debug port, PA15, PB3, 4 as normal IO

}
```

The `BSP_init()` function is the key entry point for system initialization. First, the system clock is initialized through `SystemInit()`, and then the interrupt priority group is configured as group 2. Then the delay function, serial port (USART1 is used for debugging, USART2 is used to receive infrared signals), RGB light module, and IIC communication interface for four-way motor control are initialized in turn. Finally, the multiplexing function is enabled and JTAG is remapped to SWD, and the PA15, PB3, and PB4 pins are released for use as ordinary GPIOs to ensure that all peripherals work normally and are ready to run the main application.

- bsp_motor_iic.c

```
//只能控制带编码器类型的电机    Can only control motors with encoders
//传入参数:4个电机的速度    Input parameters: speed of 4 motors
void control_speed(int16_t m1,int16_t m2 ,int16_t m3,int16_t m4)
{
    static uint8_t speed[8];
```

```

    speed[0] = (m1>>8)&0xff;
    speed[1] = (m1)&0xff;

    speed[2] = (m2>>8)&0xff;
    speed[3] = (m2)&0xff;

    speed[4] = (m3>>8)&0xff;
    speed[5] = (m3)&0xff;

    speed[6] = (m4>>8)&0xff;
    speed[7] = (m4)&0xff;

    i2cwrite(Motor_model_ADDR,SPEED_Control_REG,8,speed);

}

//控制带编码器类型的电机    Control the motor with encoder type
//传入参数:4个电机的pwm PWM of 4 motors
//此函数可以结合实时编码器的数据, 来实现control_speed的功能    This function can combine the data of
real-time encoder to realize the function of control_speed
void control_pwm(int16_t m1,int16_t m2 ,int16_t m3,int16_t m4)
{
    static uint8_t pwm[8];

    pwm[0] = (m1>>8)&0xff;
    pwm[1] = (m1)&0xff;

    pwm[2] = (m2>>8)&0xff;
    pwm[3] = (m2)&0xff;

    pwm[4] = (m3>>8)&0xff;
    pwm[5] = (m3)&0xff;

    pwm[6] = (m4>>8)&0xff;
    pwm[7] = (m4)&0xff;

    i2cwrite(Motor_model_ADDR,PWM_Control_REG,8,pwm);

}

```

control_speed: Controls the speed of the motor. It splits the incoming four motor speed values into high and low bytes, and then sends them to the motor controller via the I2C protocol to set the motor speed.

control_pwm: Controls the PWM (pulse width modulation) value of the motor. It splits the incoming four motor PWM values into high and low bytes, and sends them to the motor controller via the I2C protocol to control the speed and direction of the motor.

- app_motor.c

```

// 返回当前小车轮子轴距间距和的一半    Returns half of the current sum of the wheel axle distances
static float Motion_Get_APB(void)
{
    return Car_APB;
}

```

```

}

...

//直接控制速度    Directly control speed
void Motion_Car_Control(int16_t V_x, int16_t V_y, int16_t V_z, uint8_t adjust)
{
    float robot_APB = Motion_Get_APB();
    speed_lr = 0;
    speed_fb = V_x;
    speed_spin = (V_z / 1000.0f) * robot_APB;
    if (V_x == 0 && V_y == 0 && V_z == 0)
    {
        control_speed(0,0,0,0);
        return;
    }

    speed_L1_setup = speed_fb - speed_spin;
    speed_L2_setup = speed_fb - speed_spin;
    speed_R1_setup = speed_fb + speed_spin;
    speed_R2_setup = speed_fb + speed_spin;

    if (speed_L1_setup > 1000) speed_L1_setup = 1000;
    if (speed_L1_setup < -1000) speed_L1_setup = -1000;
    if (speed_L2_setup > 1000) speed_L2_setup = 1000;
    if (speed_L2_setup < -1000) speed_L2_setup = -1000;
    if (speed_R1_setup > 1000) speed_R1_setup = 1000;
    if (speed_R1_setup < -1000) speed_R1_setup = -1000;
    if (speed_R2_setup > 1000) speed_R2_setup = 1000;
    if (speed_R2_setup < -1000) speed_R2_setup = -1000;

    //printf("%d\t,%d\t,%d\t,%d\r\n",speed_L1_setup,speed_L2_setup,speed_R1_setup,speed_R2_setup);

    control_speed(speed_L1_setup, speed_L2_setup, speed_R1_setup, speed_R2_setup);
}

```

The `Motion_Car_Control` function is used to calculate the speed values of the four motors based on the forward speed (`V_x`), lateral speed (`V_y`) and rotation speed (`V_z`) passed in, thereby controlling the movement of the car. The function obtains the wheel axle spacing of the car through `Motion_Get_APB`, and uses this value to calculate the rotation difference (`speed_spin`) between the left and right motors, thereby adjusting the rotational movement of the car. If the speed is zero, stop the motor directly; otherwise, control the speed of the four motors of the car according to the calculated speed value to ensure that the car moves in the predetermined direction and speed.

- revaction.c

```

// 获取命令标志    Get command flag
uint8_t Get_CMD_Flag(void)

```

```

{
    return New_CMD_flag;
}

...

// 清除命令数据和相关标志 clear command data and related flags
void Clear_CMD_Flag(void)
{
    #if ENABLE_CLEAR_RXBUF
    for (uint8_t i = 0; i < New_CMD_length; i++)
    {
        RxBuffer[i] = 0;
    }
    #endif
    New_CMD_length = 0;
    New_CMD_flag = 0;
}

...

//数据分析 data analysis
void Upper_Data_Parse(uint8_t *data_buf, uint8_t num)
{
    uint8_t func_id = *(data_buf + 3);
    switch (func_id)
    {
        /* 判断功能字: 小车速度设置 Judgment function word: Trolley speed setting */
        case FUNC_MOTION:
        {
            uint8_t parm = (uint8_t) *(data_buf + 4);
            int16_t vx_recv = *(data_buf + 6) << 8 | *(data_buf + 5);
            int16_t vy_recv = *(data_buf + 8) << 8 | *(data_buf + 7);
            int16_t vz_recv = *(data_buf + 10) << 8 | *(data_buf + 9);
            uint8_t adjust = parm & 0x80;

            if (Vx_recv == 0 && Vy_recv == 0 && Vz_recv == 0)
            {
                control_pwm(0,0,0,0);
            }
            else
            {
                Motion_Car_Control(Vx_recv, Vy_recv, Vz_recv, (adjust==0?0:1));
            }
            break;
        }

        /* 判断功能字: 彩灯控制 Judgment function word: color light control */
        case FUNC_RGB:
        {
            ...
        }
    }
}

```

```

    default:
        break;
    }
}

```

The `Get_CMD_Flag` function returns the current command flag `New_CMD_flag`, which is used to check whether there is a new command; the `Clear_CMD_Flag` function clears the command flag and the data in the receiving buffer.

The `Upper_Data_Parse` function parses the data sent by the upper layer and performs different operations according to different function words (`func_id`). `FUNC_MOTION` parses and processes the speed setting command of the car, and calls `Motion_Car_Control` to control the movement or stop of the car.

- main.c

```

#define MOTOR_TYPE 2    //1:520电机 2:310电机 3:测速码盘TT电机 4:TT直流减速电机 5:L型520电机
                        //1:520 motor 2:310 motor 3:speed code disc TT motor 4:TT DC
                        reduction motor 5:L type 520 motor

int main(void)
{
    //硬件初始化 Hardware Initialization
    BSP_init();

    #if MOTOR_TYPE == 1
        ...

    #elif MOTOR_TYPE == 2
        Set_motor_type(2); //配置电机类型 Configure motor type
        delay_ms(100);
        Set_Pluse_Phase(20); //配置减速比 查电机手册得出 Configure the reduction ratio. Check the
motor manual to find out
        delay_ms(100);
        Set_Pluse_line(13); //配置磁环线 查电机手册得出 Configure the magnetic ring wire. Check the
motor manual to get the result.
        delay_ms(100);
        Set_wheel_dis(48.00); //配置轮子直径,测量得出 Configure the wheel diameter and measure
it
        delay_ms(100);
        Set_motor_deadzone(1300); //配置电机死区,实验得出 Configure the motor dead zone, and the
experiment shows
        delay_ms(100);

        ...
    #endif

    while(1)
    {
        if (Get_CMD_Flag())
        {
            Upper_Data_Parse(Get_RxBuffer(), Get_CMD_Length());
            Clear_CMD_Flag();

```

```
    }  
    }  
}
```

MOTOR_TYPE: used to set the type of motor used. Modify the corresponding number according to the comments based on the motor you are currently using.

Call the `BSP_init()` function to initialize the hardware settings, and use `Set_Motor(MOTOR_TYPE)` to set the motor type and parameters. In the `while(1)` loop, `Get_CMD_Flag` detects whether there is a new command. If so, call `Upper_Data_Parse` to parse the data and control the car to run. After processing, use `Clear_CMD_Flag` to `clear the command flag` to ensure that the system continues to respond to the host computer instructions.

4. Experimental operation

1. Burn the program to STM32.
2. Download the car drive library and PID control library in the K210\library directory to the root directory of the memory card in advance.
3. Open CanMV IDE and download the follow_apriltag.py code into the K210 module.
4. Connect all the wires of the car.
5. Place the car in a white background, move the K210 module bracket to a suitable angle, and connect the power supply.
6. Place the machine code within the camera acquisition range of the car, move the machine code, and the car will follow the machine code.

5. Experimental phenomenon

After waiting for the system to initialize, the car will not move when it does not recognize the machine code. When the machine code enters the camera acquisition range of the car, the car will start to follow the machine code and keep the machine code in the middle of the video screen. Due to the limited camera acquisition range, the machine code cannot be moved too fast, otherwise it will be out of the frame and cannot follow the machine code.

